Value Chain Physics: Formal Proof of Open Complex Giant Systems and Full-Domain Computability Based on Non-IID Dynamics

From 100-Billion Enterprise Industrial Kinetic Work Closed-Loop Empirical Proofs to First-Principle Elevation of Complexity Science

Grit Meng
Former Head & Chief Architect, Integrated Planning Solution (IPS), Lenovo Global Supply Chain
Creator of the Intelligent Planning & Control (IPC) Engine | Independent Scholar in Value Chain Physics and Systems & Complexity Science

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Abstract

Discrete manufacturing enterprises worldwide have encountered a pronounced "IT Productivity Paradox" during digital transformation: despite sustained increases in digital capital expenditures, the average Return on Invested Capital (ROIC) has deteriorated, with pervasive projects mired in localized sub-optimization and systemic instability. This paper demonstrates that the root cause lies in the long-standing reliance of conventional management and software architectures on Independent and Identically Distributed (IID) reductionist assumptions that artificially decouple procurement, manufacturing, sales, and finance. In physical value networks, material kitting, machine capacity, order commitments, and cash flows exhibit strong non-linear topological entanglement, fundamentally constituting Non-Independent and Identically Distributed (Non-IID) Open Complex Giant Systems (OCGS) as defined by Qian Xuesen. This paper establishes the analytical framework of "Value Chain Physics" grounded in Non-IID physical reality: First, it proves the Vector Cancellation Law, revealing that isolated departmental optimization inevitably triggers non-orthogonal control vector interference, dissipating energy into internal friction heat ($\Delta W_{\text{heat}} \geq 0$) and trapping the enterprise in a suboptimal Nash equilibrium. Second, it proposes a "5D Dual-Helix" architecture comprising a holographic data model and a dynamic evolution algorithm, leveraging a prior boundary operator Π and a rigid manifold $\Pi_{\perp}$ to strip normal friction degrees of freedom, algebraically pruning factorial phase space search complexity from $\mathcal{O}(N!)$ to an empirical polynomial bound $\mathcal{O}(N\log N)$. Third, it introduces a compliance compact support projection operator E_{supp} (Conscience Operator) to cap residual variance against Goodhart breakdown, proving polynomial closed-loop computability ($T(N) \leq \tau_{\text{phy}} < +\infty$) and convergence to a unique absolute optimal steady state under physical instability window truncation. On this foundation, the paper provides rigorous mathematical formal proofs for Qian Xuesen's three foundational OCGS principles: proving the Necessity of the Overall Design Department for breaking Nash deadlocks and its Sufficiency when coupled with rigid barriers; the Sufficiency of the Hall for Workshop of Metasynthetic Engineering (HWME) for closed-loop computability under causal orthogonal pruning; and the Necessity of the Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} for breaking silicon internal closure alongside the Sufficiency of the work operator tensor product $\Phi_{\text{Human}} \otimes \mathcal{A}_{\text{Silicon}}$ in human-machine combination. This work supplies mathematical and cybernetic foundations for autonomous self-healing closed-loop control and ROIC elevation in physical enterprise value networks.

Keywords: Value Chain Physics; Non-Independent and Identically Distributed (Non-IID); Open Complex Giant Systems (OCGS); Vector Cancellation; 5D Dual-Helix; Closed-Loop Computability;

1. Introduction: From Non-IID Physical Reality to Value Chain Physics

In modern enterprise management and industrial engineering, Return on Invested Capital (ROIC) serves as the core physical metric for evaluating enterprise operational efficiency and capital allocation capability:

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested Capital}} = \frac{\text{Net Operating Profit After Tax (NOPAT)}}{\text{Invested Capital (Fixed Assets + Working Capital)}}$$

However, over the past decade, the compound annual growth rate of digital investment in Chinese and global discrete manufacturing enterprises has exceeded 15%. Systems such as Enterprise Resource Planning (ERP), Advanced Planning and Scheduling (APS), Manufacturing Execution Systems (MES), Warehouse Management Systems (WMS), Supplier Relationship Management (SRM), and Supply Chain Control Towers have been deployed at scale. In stark contrast, the average manufacturing ROIC has dropped from 8% to less than 5%, with approximately 70% of digital projects failing to meet expectations; even projects evaluated as "successful" have generally failed to drive a substantive improvement in ROIC. This constitutes the vivid manifestation of the "IT Productivity Paradox" proposed by Brynjolfsson (1993) in the era of industrial intelligence.

Existing literature often attributes this to insufficient management execution, organizational silos, or data quality flaws. However, this paper argues that the root cause of this paradox lies in the paradigm fracture between value chain management and underlying system science:

- **The Static Assumption of IID**: Since Taylorism and modern modular enterprise architectures (e.g., 4A architecture, Business Process Management BPM, Sales and Operations Planning S&OP, Balanced Scorecard BSC) were established, management science has split enterprises into independent functions such as procurement, manufacturing, sales, and finance, assuming that state evolution across departments follows Independent and Identically Distributed (IID) dynamics.
- **Non-IID Physical Reality and the OCGS Challenge**: In real discrete manufacturing value networks, material kitting, equipment capacity, order delivery, and financial cash flows exhibit extremely strong non-linear topological entanglement (Cao, 2014, 2022). The system is fundamentally a Non-Independent and Identically Distributed (Non-IID) Open Complex Giant System (OCGS), as defined by Qian Xuesen.

Navigating a Non-IID strongly coupled 3D physical world using IID 2D independent tools inevitably causes control vectors across departments—driven by local Key Performance Indicators (KPIs)—to interfere in non-orthogonal directions. This generates dynamic vector cancellation and friction heat (ΔW_{heat}), locking the enterprise into a low-efficiency Nash equilibrium.

To break through this dilemma, this paper proposes "Value Chain Physics"—as the concrete projection of systems science in physical value networks. Adhering to the first principles of systems science, this paper begins with the absolute physical reality of Non-IID dynamics to establish a rigorous Value Chain Physics analytical system; while deriving Value Chain Physics, it formally proves the correctness of Qian Xuesen's Open Complex Giant System theory projected into value chain management, supplying mathematical formal proofs for Qian Xuesen's three core principles of complex giant system governance.

2. Theoretical Foundations: Value Chain Physics and Qian Xuesen's Open Complex Giant Systems (OCGS)

This study stands upon the intersection and elevation of three theoretical pillars:

2.1 Qian Xuesen's Open Complex Giant System (OCGS) Theory: From Qualitative Description to Value Chain Physics Projection

Qian et al. (1990) defined man-made and natural systems featuring massive elements, multi-level structures, long-range non-linear interactions, and continuous exchange of matter, energy, and information with the environment as Open Complex Giant Systems (OCGS). Qian proposed the "Hall for Workshop of Metasynthetic Engineering (HWME) from qualitative to quantitative", emphasizing the systems engineering philosophy of "Overall Design Department" and "Human-Machine Combination."

However, Qian Xuesen's theory has long remained at macro-qualitative and methodological levels in traditional management and software engineering, lacking formal mathematical derivations and algorithmic closed loops in concrete physical domains (such as discrete manufacturing value networks). Value Chain Physics is precisely the dimensional-reduction projection and specification of systems science in discrete manufacturing and value network phase spaces.

2.2 Professor Longbing Cao's Non-IID Theory and Isomorphism with Value Chain Physics

Professor Longbing Cao (2014, 2022)—a disciple of Academician Ruwei Dai at CAS, who in turn studied under Qian Xuesen—systematically pioneered Non-IID Learning theory, proving that real-world systems universally feature intra-entity coupling, inter-entity interaction coupling, and spatiotemporal heterogeneity. Professor Cao proved a core mathematical theorem: "If a system is fundamentally Non-IID, modeling and solving it under IID assumptions produces results that are mathematically biased and incorrect."

According to the Holographic Anti-Entropy Theoretical System (Meng, 2026), a strict fourfold topological isomorphism exists between Holographic Anti-Entropy theory and Professor Longbing Cao's Non-IID theory:

1. **Diagnosis Isomorphism**: Non-IID strong coupling ⇔ Non-orthogonal topological interference across nodes triggering factorial complexity $\mathcal{O}(N!)$;
2. **Solution Path Isomorphism**: High-dimensional coupling feature extraction ⇔ Prior boundary operator Π dimensional reduction and rigid manifold $\Pi_{\perp}$ normal clipping, collectively named 5D Dual-Helix algebraic pruning ($\mathcal{O}(N!) \to \mathcal{O}(N\log N)$);
3. **Safety Defense Isomorphism**: Hacker attacks and model bias ⇔ Compliance domain compact support projection operator E_{supp} clipping heavy-tailed residual distributions;
4. **Ultimate Closed-Loop Isomorphism**: Human-in-the-loop decision support ⇔ Decision write-back automation and Human-out-of-the-loop (HOOTL) self-healing closed loops.

2.3 Cybernetics and Information Theory Bounds

Ashby's (1956) Law of Requisite Variety requires the controller's variety V_c to be no less than the controlled system's variety V_s; meanwhile, Miller's (1956) Law constrains human carbon-based working memory bandwidth (7 ± 2 chunks). When system variety greatly exceeds controller capacity ($V_s > V_c$), traditional open-loop manual coordination inevitably breaks down. Kalman's (1960)

Controllability-Observability Duality Theorem further demonstrates that if underlying physical states are unobservable, top-level intent is uncontrollable.

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3. Value Chain Physics: 5D Complete State Manifold and Eight Architecture Design Principles

3.1 Physical 5D Complete State Manifold (SCI Topological Invariants)

In Value Chain Physics, the physical reality of a value network at time t is formally expressed as a Systemically Complete and Irreducible (SCI) 5D topological manifold:

$$\mathcal{M}_{\text{model}} = \langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$$

- $\mathcal{N}$ **(Nodes)**: Micro-entity set of all in-boundary physical machinery, materials, and control units;
- $\mathcal{T}$ **(Topology)**: Non-linear connectivity matrix $\boldsymbol{A}$ and Directed Acyclic Graph (DAG) among nodes;
- $\mathcal{C}$ **(Constraint Bundles)**: Degree-of-freedom limits C imposed by rigid manifold $\Pi_{\perp}$ (capacity ceilings, material lead times, rigid deadlines);
- $X(t)$ **(State Vectors)**: Instantaneous state coordinate vector $X(t)$ in state space Ω;
- $\Delta X(t)$ **(State Transitions)**: Phase transition and evolution path passively triggered by residual $\Delta = X_{\text{real}}(t) - X_{\text{target}}(t)$.

According to Holographic Anti-Entropy theory, the 5D foundational primitives—Nodes, Topology, Constraint Bundles, State Vectors, State Transitions—constitute Systemically Complete and Irreducible (SCI) topological invariants. Stripping any dimension causes manifold degradation, losing the capacity to execute anti-entropy work against the environment. In raw phase spaces, significant non-linear coupling exists among the 5 dimensions; non-orthogonal interference is the physical origin of vector cancellation. After prior boundary operator Π dimensional reduction and rigid manifold $\Pi_{\perp}$ clipping, orthogonal projection over controllable sub-spaces yields an orthogonal structure within feasible space Ω_{feasible}.

[Causal Anchoring and Dual-Duality Decision Practice in Action]

1. **Fourth and Fifth Dimension Reality**: Fourth dimension state vector X is the holographic production data row (containing baseline initial plans, previous plan states, and latest actual warehouse/WIP execution states); fifth dimension state transition ΔX is the Causal Anchoring chain and residuals with causality;
2. **PO / IC / WIP / FG Causal Anchoring**: Purchase Orders (PO), In-Stock Inventory (IC), Machine Work-In-Progress (WIP) are rigidly anchored to Sales Orders (SO) and Finished Goods (FG) through concrete manufacturing and logistics work;
3. **Reverse Traceback Judgment along Causal Anchoring Chains**:
    - **The World is Off**: Truck delayed on highway by 2 hours, supplier lead time extended → 1st-order schedule adjustment pulls self-healing;
    - **The Model is Wrong**: Geopolitical conflict blocks sea ports, critical machinery physically scrapped → activate Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} to reconstruct network topologies.

If the world is off, adjust execution; if the model is wrong, apply second-order amendment.

3.2 Eight Architecture Design Principles of Value Networks

Starting from the first principle of Non-IID strongly coupled state spaces, Value Chain Physics formulates Eight Architecture Design Principles governing value network evolution:

- **Principle 1 Teleological (Governance Principle)**: World reality is fundamentally Non-IID. Lacking global orchestration, local self-interested optimization triggers non-orthogonal vector interference, generating dissipative friction heat ΔW_{heat} and locking the system into a low-efficiency Nash equilibrium.
- **Principle 2 Ontological (Complexity Boundary Principle)**: Under Non-IID coupling, unconstrained solution spaces explode factorially $\mathcal{O}(N!)$, exceeding carbon-based working memory. High-frequency micro-computation must be yielded to silicon engines.
- **Principle 3 Schematic (5D Dual-Helix Principle)**: Dimensional reduction governance must rely on the 5D Dual-Helix of "Holographic Data Model (Left Helix) + Dynamic Evolution Algorithm (Right Helix)," executing algebraic pruning on rigid manifold $\Pi_{\perp}$.
- **Principle 4 Capability (Tripartite Fusion Principle)**: Business ontology is indivisible; capability building must fuse "Business Decoding (penetrating coupling), System Modeling (solidifying coupling), Closed-Loop Collaboration (harnessing coupling)" into a tripartite unity.
- **Principle 5 Mechanism (Quota Autonomy and Central Coordination Principle)**: Centralized prior boundary drawing and rigid barrier governance outperform pure decentralized negotiation, guiding sub-domain autonomy by adjusting global quota vectors $\boldsymbol{r} \in \mathcal{C}$.
- **Principle 6 Path (Observability and Decision Write-Back Principle)**: Control domain dimensions are bounded by observation domain dimensions ($\dim\mathcal{C} \leq \dim\mathcal{O}$); control mechanisms must achieve "Human-out-of-the-loop, decision automated write-back" to physical nodes.
- **Principle 7 Dynamics (Physical Fulcrum and Leverage Principle)**: Control energy must act precisely upon tightly constrained bottleneck fulcrums $X_{\text{fulcrum}} \in \mathcal{C}_{\text{active}}$, closing mismatch angles ($\theta \to 0 \Rightarrow \cos\theta \to 1$) to maximize effective work.
- **Principle 8 Evolutionary (Second-Order Meta-Cognitive Constitutional Rewriting Principle)**: Normal operations self-heal autonomously via silicon out-of-the-loop; when residuals breach thresholds $\Delta > \theta_{\text{trigger}}$ triggering phase deadlocks, human experts introduce the Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} (Meta-cognitive Constitutional Rewriting Operator, corresponding to high-level prefrontal self-reflection in cognitive science) to rewrite axiomatic bases ($\Phi: \Pi_k \to \Pi_{k+1}$).

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4. Formal Derivations: Vector Cancellation, Dual-Helix Pruning, and Formal Proof of Qian Xuesen's OCGS Core Principles

This section begins with Non-IID physical reality and Value Chain Physics operators to unfold quantitative derivations and supply formal mathematical proofs for Qian Xuesen's three core OCGS principles.

4.1 Theorem 1: Vector Cancellation Law and Physical Waste Heat Theorem

In a Non-IID value network, let N micro-nodes/departments exert control and intent vectors $\boldsymbol{v}_i = \frac{dX_i}{dt}$. When non-orthogonal coupling and mutual gaming exist among nodes, by the vector triangle inequality:

$$\left\| \sum_{i=1}^{N} \boldsymbol{v}_i \right\| \leq \sum_{i=1}^{N} \| \boldsymbol{v}_i \|$$

The control work energy canceled out due to physical interference is defined as internal Coordinated Dissipative Heat:

$$\Delta W_{\text{heat}} \triangleq \sum_{i=1}^{N} \| \boldsymbol{v}_i \| - \left\| \sum_{i=1}^{N} \boldsymbol{v}_i \right\| \geq 0$$

Physically, ΔW_{heat} manifests as non-orthogonal control vector interference; operationally, this difference translates directly into extended Work-In-Progress (WIP) durations and increased idle man-hours; financially, it causes Cost of Goods Sold (COGS) to rise and Net Working Capital (NWC) to be trapped, reducing NOPAT, inflating Invested Capital, and driving ROIC downward. When mismatch angle $\theta \to 0$, $\cos\theta \to 1$, effective work efficiency reaches its physical limit $W_{\text{eff}} = W_{\text{total}} \cdot \cos\theta \to W_{\text{total}}$.

4.2 Theorem 2: 5D Dual-Helix Algebraic Pruning and Complexity Convergence

Dimensional reduction governance operates through the Dual-Helix structure:

- **Left Helix (Holographic Data Model Helix)**: Maps physical 5D into rigid manifold $\Pi_{\perp} = (\mathcal{N}, \mathcal{T}, \mathcal{C})$, loading $X(t)$ and historical causality $\Delta X(t-1)$;
- **Right Helix (Dynamic Evolution Algorithm Helix)**: Algorithm $\mathcal{A}$ executes a recursive loop of "concurrent pre-computation → manifold evaluation → redundant degree-of-freedom algebraic pruning."

The system equation is expressed as the pre-computation, in-computation, and post-computation recurrence triplet:

$$\begin{cases} \mathcal{M}(t) = \{\Pi_{\perp}, X_{\text{target}}(t), X_{\text{state}}(t), \Delta X(t-1)\} \\ \quad \text{(Pre-Computation: Holographic Loading of 5D State)} \\ (u^*(t), \Delta X(t)) = \mathcal{A}(\mathcal{M}(t)) \\ \quad \text{(In-Computation: Solving Controls and Transitions)} \\ X(t+1) = X(t) + \Delta X(t) \\ \quad \text{(Post-Computation: Advancing to Next State)} \end{cases}$$

Proof: Let N denote the total number of instantiated demand-BOM-routing nodes in the manufacturing network. Prior boundary operator Π prunes the raw $\mathcal{O}(N!)$ permutation phase space into a finite feasible space Ω_{feasible} via topological sorting and constraint propagation; rigid manifold $\Pi_{\perp}$ strips normal friction degrees of freedom ($x_{\perp}(t) \to 0$). Unconstrained permutation bounds explode factorially $\mathcal{O}(N!)$; prior operator Π fixes causal order along a Directed Acyclic Graph (DAG), reducing topological sorting complexity to $\mathcal{O}(N+E)$.

Because physical BOM fan-in per assembly node is bounded by a finite engineering constant and shared sub-assemblies merge as single DAG nodes rather than expanding into duplicate trees, $E =$

$\mathcal{O}(N)$ and Stage 1 additive material kitting propagation drops to linear complexity $\mathcal{O}(N)$. Furthermore, because extensive materials can be stored across periods, any order 100% kitted at its earliest material availability date t_{avail} remains 100% kitted across every subsequent slot $t' \in [t_{\text{avail}}, T_{\text{commit}}]$; thus Stage 2 local sorting and matching of intensive machine capacities within pruned windows never invalidates Stage 1 kitting, eliminating cross-stage backtracking and yielding a tight polynomial bound of $\mathcal{O}(N\log N)$. In general Non-IID networks without causal DAG structure, worst-case complexity degrades to $\mathcal{O}(b^L)$. Thus, $\mathcal{O}(N\log N)$ represents the tight polynomial upper bound for causal DAG manufacturing networks under $\Pi_\perp$ orthogonal pruning, conquering the combinatorial explosion of complex manufacturing systems.

4.3 Theorem 3: Causal DAG Finite Truncation Closed-Loop Convergence and Compact Support Projection

In physical phase space, clock step $\Delta t_{\text{clock}} > 0$ and instability window $\tau_{\text{phy}} < +\infty$ truncate causal DAG evolution depth to finite steps $K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}} < +\infty$. The 5D Dual-Helix explicitly strips non-orthogonal branches inside rigid manifold $\Pi_\perp$, eliminating time-reversal and causal-backtracking spurious degrees of freedom. For a fixed boundary, single-step orthogonal matching and pruning complexity is $O(N\log N)$. Advancing along unidirectional causal lines for K steps, worst-case computation cost is strictly bounded:

$$T(N) = O(K \cdot N\log N) \in \boldsymbol{P}, \quad \text{and } T(N) \leq \tau_{\text{phy}} < +\infty$$

For any admissible closed-loop trajectory $S \in \Omega_{\text{feasible}}(\mathcal{C}_k)$ that walks through both hard physical constraints $\mathcal{C}_{\text{hard}}$ and agreed order commitment targets $\mathcal{C}_{\text{soft}}$ (excluding trivial idle zero-work $\boldsymbol{v}_i = \mathbf{0}$), total value-chain loss decomposes into $W_{\text{loss}}^{\text{total}}(S) = W_{\text{intrinsic}}(\mathcal{C}_{\text{hard}}) + \Delta W_{\text{heat}}(S)$, where $W_{\text{intrinsic}}(\mathcal{C}_{\text{hard}})$ is the irreducible physical floor dictated by hard material lead times and machine cycle limits, and $\Delta W_{\text{heat}}(S) \geq 0$ is internal vector-cancellation waste heat. Because constraint manifold $\Pi_\perp$ enforces $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$, successful traversal in finite steps yields $\Delta W_{\text{heat}}(S_k^*) = 0$ and $W_{\text{loss}}^{\text{total}}(S_k^*) = W_{\text{intrinsic}}(\mathcal{C}_{\text{hard}})$. Any hypothetical feasible trajectory S' with strictly lower loss $W_{\text{loss}}^{\text{total}}(S') < W_{\text{loss}}^{\text{total}}(S_k^*)$ would require $\Delta W_{\text{heat}}(S') < 0$, contradicting the vector triangle inequality $\Delta W_{\text{heat}} \geq 0$. Combining spatiotemporal machine exclusivity ($\rho_i \leq \rho_{\max}$) with a strict lexicographic tie-breaking order $(t_{\text{avail}}, T_{\text{commit}}, -\mathcal{P}_{\text{biz}}, \text{ID})$ for simultaneous equal-urgency orders eliminates symmetric permutation ties, guaranteeing deterministic convergence to a unique absolute optimal steady state S_k^* ('Constraint Walk-Through ∧ Zero Cancellation ⇔ In-Boundary Global Optimality and Uniqueness $\exists! S_k^*$').

To prevent Goodhart's Law breakdown under extreme optimization pressure (when an indicator is chosen as a performance target, it ceases to be a good indicator and induces extreme arbitrage, Goodhart, 1984; $\lim_{\text{optimization}\to\infty} \mathbb{E}(r^*) = -\infty$), we introduce the compliance domain compact support projection operator E_{supp} (Conscience Operator), suspended on target and residual distributions. Operator E_{supp} acts upon residual probability density $p(\Delta)$, truncating residual support to a compact deadband $[-\theta_{\text{dead}}, \theta_{\text{dead}}]$. When residuals attempt to breach boundaries, they are physically truncated, strictly capping residual variance:

$$\text{Var}(\Delta X) \leq K_{\text{supp}} < \infty$$

blocking system state divergence risks caused by target distortion.

4.4 Core Theorem: Formal Proof Theorem of Qian Xuesen's OCGS Three Principles

> **Prerequisite**: Under observer prior operator Π boundary drawing, rigid manifold $\Pi_{\perp}$ constraint, dissipation mechanisms, and Φ self-reflection constructing $K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}} < +\infty$.

This paper formally proposes and proves the formal proof theorem for Qian Xuesen's three core OCGS principles in entity discrete manufacturing value networks as a class of OCGS featuring causal DAG structures:

Theorem (Formal Proof Theorem for Qian Xuesen's Overall Design Department, HWME, and Human-Machine Combination/Human-Led Principles):
Let any Open Complex Giant System $\mathcal{S}_{\text{OCGS}}$ exist under Non-IID strongly coupled dynamics:

1. **Mathematical Necessity and Sufficiency Proof for "Overall Design Department"**: Proving that under Non-IID dynamics, lacking unified prior boundary operator Π and centralized quota/manifold governance forces sub-domain optimizations into vector triangle inequalities, generating global vector cancellation waste heat $\Delta W_{\text{heat}} = \sum \| \boldsymbol{v}_i \| - \| \sum \boldsymbol{v}_i \| > 0$ and locking the system into non-cooperative Nash equilibrium deadlocks and Brownian motion internal friction. Thus, under our Non-IID vector cancellation framework, Qian Xuesen's "Overall Design Department" is the Necessary condition for breaking Nash deadlocks and eliminating waste heat; coupled with linear manifold constraints imposed by rigid barrier $\Pi_{\perp}$, it constitutes the Sufficient condition for multi-dimensional coordination.
2. **Closed-Loop Computability Proof for "Hall for Workshop of Metasynthetic Engineering (HWME)"**:
   Proving that the 5D Dual-Helix (Holographic Data Model + Dynamic Evolution Algorithm) prunes phase space to Ω_{feasible} via prior operator Π, strips normal degrees of freedom ($x_{\perp}(t) \to 0$) on rigid manifold $\Pi_{\perp}$, and compresses factorial search spaces $\mathcal{O}(N!)$ into empirical polynomial complexity $\mathcal{O}(N \log N)$ (or general branch-and-bound upper bound $\mathcal{O}(b^L)$) via topological sorting and constraint propagation. Under observer Π boundary drawing, $\Pi_{\perp}$ constraint, and physical instability window truncation ($K < +\infty$), causal decoupling polynomial closed-loop computability analytically guarantees the existence and closed-loop convergence of a unique optimal steady state S_k^* (whose work efficiency strictly surpasses any open-loop Nash equilibrium). This constitutes the mathematical mechanism by which Qian's HWME conquers computational irreducibility, supplying the Sufficient mechanism for closed-loop computability; necessity is defined by complexity lower bounds: general Non-IID scheduling networks (including Job-Shop Scheduling as a special case) are NP-hard, so no universal polynomial-time algorithm exists for all instances; our framework guarantees polynomial closed-loop computability based on physical instability window truncation of causal DAG depth ($K < \infty$).
3. **Necessity and Sufficiency Proof for "Human-Machine Combination / Human-Led"**:
   Pure silicon first-order algorithm $\mathcal{A}_{\text{Silicon}}$ is a deterministic closed loop inside given prior boundaries Π_k and fixed axiom/constraint sets. By formal system internal closure and non-self-bootstrapping theorems, algorithms cannot self-sufficiently prove or replace their own axioms internally; when residuals $\Delta > \theta_{\text{trigger}}$ breach the expressive range of Π_k, $\mathcal{A}$ alone inside Π_k becomes unsolvable or suffers reconstruction failure. Second-Order Meta-Cognitive Constitutional Rewriting Operator $\Phi_{\text{Human}}: \Pi_k \to \Pi_{k+1}$ (corresponding to high-level prefrontal self-reflection in cognitive science) provides cross-framework axiom rewriting capacity. The global work operator must be expressed as:

$$\text{Global Work Operator} = \Phi_{\text{Human}} \otimes \mathcal{A}_{\text{Silicon}}$$

Within our operator decomposition framework, Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} is the Necessary condition for breaking silicon formal system internal

closure; work operator $\Phi_{\text{Human}} \otimes \mathcal{A}_{\text{Silicon}}$ is the Sufficient condition for achieving generational system evolution. If $\Phi \equiv 0$ (pure silicon closed loop), when residuals breach critical thresholds, no analytical self-healing trajectory exists within given axiom sets. Generalization to general OCGS requires reconstructing the physical carrier of Φ and observation domains.

Proof Summary:

1. **Overall Design Department Necessity & Sufficiency**: Under Non-IID connectivity matrix $\boldsymbol{A}$, independent sub-node optimization yields negative vector dot products ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle < 0$). By vector triangle inequalities, composite vector magnitude satisfies $\|\sum \boldsymbol{v}_i\| < \sum \| \boldsymbol{v}_i \|$, inevitably generating $\Delta W_{\text{heat}} > 0$. In our linear vector cancellation framework, an Overall Design Department is Necessary for breaking Nash deadlocks; coupled with $\Pi_{\perp}$ linear manifold constraints, it is Sufficient for multi-dimensional coordination.
2. **HWME Closed-Loop Computability**: Via Π boundary drawing and topological sorting, permutation space $N!$ converges to causal topological order on DAGs, reducing refinement complexity to $\mathcal{O}(N \log N)$; after rigid manifold $\Pi_{\perp}$ strips normal components, internal friction dissipation and residual-triggered Φ rewriting jointly drive network truncation depth $K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}} < +\infty$. By polynomial closed-loop computability under instability time truncation, iterative sequences form unidirectional forward deductions without backtracking during fixed-structure stages, guaranteeing convergence to a unique steady state S_k^* (strictly superior to any open-loop Nash equilibrium). Under boundary and dissipation constructs, closed-loop computability holds.
3. **Human-Machine Operator Necessity & Sufficiency**: Let first-order algorithm system be $\mathcal{A}$ with state space bounded by finite axiom set Π_k. By formal system non-self-bootstrapping, for any $\Delta > \theta_{\text{trigger}}$, no analytical self-healing trajectory exists inside Π_k (first-order algorithms cannot self-rewrite their own axioms). Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} possesses second-order hyper-formal mapping $\Phi: \text{Map}(\Xi \to \Omega_k) \to \text{Map}(\Xi \to \Omega_{k+1})$. Thus, Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} is Necessary for breaking closure, while its tensor product with silicon computing power $\mathcal{A}_{\text{Silicon}}$ is Sufficient for generational self-healing evolution.

Q.E.D.

Under our framework (Π boundary drawing, $\Pi_{\perp}$ constraint, dissipation construct $K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}} < +\infty$, Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human}, compliance support operator (Conscience Operator) E_{supp} redline), for physical value network OCGS, this paper supplies the Necessary condition for the Overall Design Department to break Nash deadlocks and the Sufficient condition when coupled with rigid barriers; supplies the Sufficient condition for HWME closed-loop computability on restricted graphs; and supplies the Necessary condition for Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} and the Sufficient condition for work operator tensor products in human-machine combination. Generalization to general OCGS requires reconstructing Π, matrix $\boldsymbol{A}$, operator Φ, and observation domains for specific domains.

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5. Architecture Governance: “Tripartite” Capabilities and Medical School Delivery Model

Addressing Non-IID physical reality and OCGS characteristics, this paper proposes three essential architectural governance pillars:

- **“Human-out-of-the-loop, Human-machine collaboration” Division**: Humans manage Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} and compliance redlines (compliance support operator / Conscience Operator E_{supp}); silicon engines execute high-speed first-order algebraic pruning and automated decision write-back inside rigid manifold $\Pi_{\perp}$.
- **“Tripartite” Fusion Architecture Capabilities**:
    - **Business Decoding Capability**: Penetrating organizational silos to parse fine-grained strongly coupled causal topologies of physical entities;
    - **System Modeling Capability**: Losslessly mapping decoded logic into 5D complete state manifolds and convergent algorithms;
    - **Closed-Loop Collaboration Capability**: Achieving automated decision write-back and residual self-healing.
- **Delivery Sovereignty’s “Medical School Model”**: Departing from traditional sell-and-forget software packages or black-box subscriptions, the “Medical School Model” supplies Open-Box source code delivery, dual team cultivation (business + algorithms), and long-term self-healing sovereignty, empowering enterprises to build self-sustaining anti-entropy capacities.

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6. Conclusion

The fundamental cause of ROIC degradation during digital transformation lies in solving Non-IID strongly coupled physical reality using IID weakly coupled static assumptions. Bridging value chain management and system science, this paper begins with Non-IID physical reality to establish the Value Chain Physics paradigm, deriving the 5D complete state manifold and Eight Architecture Design Principles; and formally proving the Vector Cancellation Law, 5D Dual-Helix algebraic pruning, and causal DAG unidirectional evolution convergence theorems.

The core theoretical breakthrough of this paper lies in supplying formal mathematical proofs for Qian Xuesen’s three core OCGS principles (Necessary condition for the Overall Design Department to break Nash deadlocks and Sufficient condition when coupled with rigid barriers; Sufficient condition for HWME closed-loop computability on restricted graphs; Necessary condition for Second-Order Meta-Cognitive Constitutional Rewriting Operator Φ_{Human} and Sufficient condition for work operator tensor products in human-machine combination) in an entity discrete manufacturing value network as a class of OCGS featuring causal characteristic line orthogonal pruning, bypassing Wolfram computational irreducibility hurdles under restricted structures. Constructing physical “Plan-Execute-Feedback” self-healing closed loops is the effective path for eliminating internal vector cancellation and elevating ROIC.

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